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Dataset Validation Report
Dataset: datasets\synthetic\synthetic_dataset_v2.csv

missing_values: fail
  - missing_by_column: {'career': 0, 'years_experience': 0, 'education_level': 20000, 'projects_completed': 0, 'certifications': 0, 'python_score': 0, 'java_score': 0, 'javascript_score': 0, 'sql_score': 0, 'machine_learning_score': 0, 'deep_learning_score': 0, 'cloud_score': 0, 'devops_score': 0, 'cybersecurity_score': 0, 'data_analysis_score': 0, 'database_score': 0, 'networking_score': 0, 'mobile_score': 0, 'game_dev_score': 0, 'testing_score': 0, 'business_analysis_score': 0, 'product_management_score': 0, 'ui_design_score': 0, 'ux_research_score': 0, 'communication_score': 0, 'leadership_score': 0, 'problem_solving_score': 0, 'teamwork_score': 0, 'agile_score': 0, 'research_score': 0, 'salary_band': 0, 'remote_preference': 20000, 'career_growth_score': 0, 'job_satisfaction': 0, 'work_hours_per_week': 0, 'country': 20000, 'industry': 20000, 'employment_type': 20000}
  - missing_rows: 20000

duplicate_rows: pass
  - duplicate_count: 0

class_balance: pass
  - counts: {'Data Scientist': 1000, 'ML Engineer': 1000, 'Database Administrator': 1000, 'Backend Developer': 1000, 'Network Engineer': 1000, 'UI/UX Designer': 1000, 'Cloud Engineer': 1000, 'AI Research Engineer': 1000, 'Game Developer': 1000, 'Product Manager': 1000, 'Ethical Hacker': 1000, 'Full Stack Developer': 1000, 'Mobile App Developer': 1000, 'Software Engineer': 1000, 'QA Engineer': 1000, 'Business Analyst': 1000, 'Data Engineer': 1000, 'Cyber Security Analyst': 1000, 'DevOps Engineer': 1000, 'Frontend Developer': 1000}
  - balance_ratio: 0.05

career_balance: pass
  - expected_per_career: 1000
  - counts: {'Software Engineer': 1000, 'Backend Developer': 1000, 'Frontend Developer': 1000, 'Full Stack Developer': 1000, 'Mobile App Developer': 1000, 'Data Scientist': 1000, 'ML Engineer': 1000, 'Data Engineer': 1000, 'AI Research Engineer': 1000, 'DevOps Engineer': 1000, 'Cloud Engineer': 1000, 'Cyber Security Analyst': 1000, 'Ethical Hacker': 1000, 'Network Engineer': 1000, 'UI/UX Designer': 1000, 'Product Manager': 1000, 'Business Analyst': 1000, 'QA Engineer': 1000, 'Database Administrator': 1000, 'Game Developer': 1000}
  - deviations: {'Software Engineer': 0, 'Backend Developer': 0, 'Frontend Developer': 0, 'Full Stack Developer': 0, 'Mobile App Developer': 0, 'Data Scientist': 0, 'ML Engineer': 0, 'Data Engineer': 0, 'AI Research Engineer': 0, 'DevOps Engineer': 0, 'Cloud Engineer': 0, 'Cyber Security Analyst': 0, 'Ethical Hacker': 0, 'Network Engineer': 0, 'UI/UX Designer': 0, 'Product Manager': 0, 'Business Analyst': 0, 'QA Engineer': 0, 'Database Administrator': 0, 'Game Developer': 0}

feature_ranges: fail
  - issues: [{'column': 'years_experience', 'expected_range': [1, 12], 'out_of_range_rows': 5457}, {'column': 'projects_completed', 'expected_range': [3, 25], 'out_of_range_rows': 4798}, {'column': 'certifications', 'expected_range': [0, 4], 'out_of_range_rows': 1861}, {'column': 'python_score', 'expected_range': [4, 8], 'out_of_range_rows': 1591}, {'column': 'java_score', 'expected_range': [3, 8], 'out_of_range_rows': 287}, {'column': 'javascript_score', 'expected_range': [5, 9], 'out_of_range_rows': 6343}, {'column': 'sql_score', 'expected_range': [5, 9], 'out_of_range_rows': 6234}, {'column': 'machine_learning_score', 'expected_range': [1, 5], 'out_of_range_rows': 7594}, {'column': 'deep_learning_score', 'expected_range': [1, 3], 'out_of_range_rows': 18193}, {'column': 'cloud_score', 'expected_range': [3, 8], 'out_of_range_rows': 263}, {'column': 'devops_score', 'expected_range': [2, 7], 'out_of_range_rows': 281}, {'column': 'cybersecurity_score', 'expected_range': [1, 5], 'out_of_range_rows': 6505}, {'column': 'data_analysis_score', 'expected_range': [3, 7], 'out_of_range_rows': 523}, {'column': 'database_score', 'expected_range': [4, 8], 'out_of_range_rows': 1861}, {'column': 'networking_score', 'expected_range': [2, 6], 'out_of_range_rows': 1925}, {'column': 'testing_score', 'expected_range': [4, 8], 'out_of_range_rows': 1860}, {'column': 'business_analysis_score', 'expected_range': [2, 6], 'out_of_range_rows': 1965}, {'column': 'product_management_score', 'expected_range': [1, 5], 'out_of_range_rows': 6553}, {'column': 'ui_design_score', 'expected_range': [2, 5], 'out_of_range_rows': 6940}, {'column': 'ux_research_score', 'expected_range': [1, 4], 'out_of_range_rows': 13504}, {'column': 'communication_score', 'expected_range': [6, 9], 'out_of_range_rows': 13190}, {'column': 'leadership_score', 'expected_range': [3, 8], 'out_of_range_rows': 272}, {'column': 'problem_solving_score', 'expected_range': [7, 10], 'out_of_range_rows': 18221}, {'column': 'teamwork_score', 'expected_range': [6, 9], 'out_of_range_rows': 13505}, {'column': 'agile_score', 'expected_range': [4, 8], 'out_of_range_rows': 1785}, {'column': 'research_score', 'expected_range': [3, 7], 'out_of_range_rows': 491}, {'column': 'salary_band', 'expected_range': [85000, 160000], 'out_of_range_rows': 4877}]

impossible_combinations: pass
  - issues: []

outliers: fail
  - outliers: [{'column': 'python_score', 'count': 4, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'java_score', 'count': 1, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'javascript_score', 'count': 1, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'machine_learning_score', 'count': 2, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'deep_learning_score', 'count': 2, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'devops_score', 'count': 1, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'cybersecurity_score', 'count': 1, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'database_score', 'count': 2, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'business_analysis_score', 'count': 2, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'ui_design_score', 'count': 1, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'leadership_score', 'count': 1, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'problem_solving_score', 'count': 1, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'research_score', 'count': 2, 'lower_bound': 1.0, 'upper_bound': 9.0}, {'column': 'salary_band', 'count': 209, 'lower_bound': 39327.38, 'upper_bound': 222344.38}]
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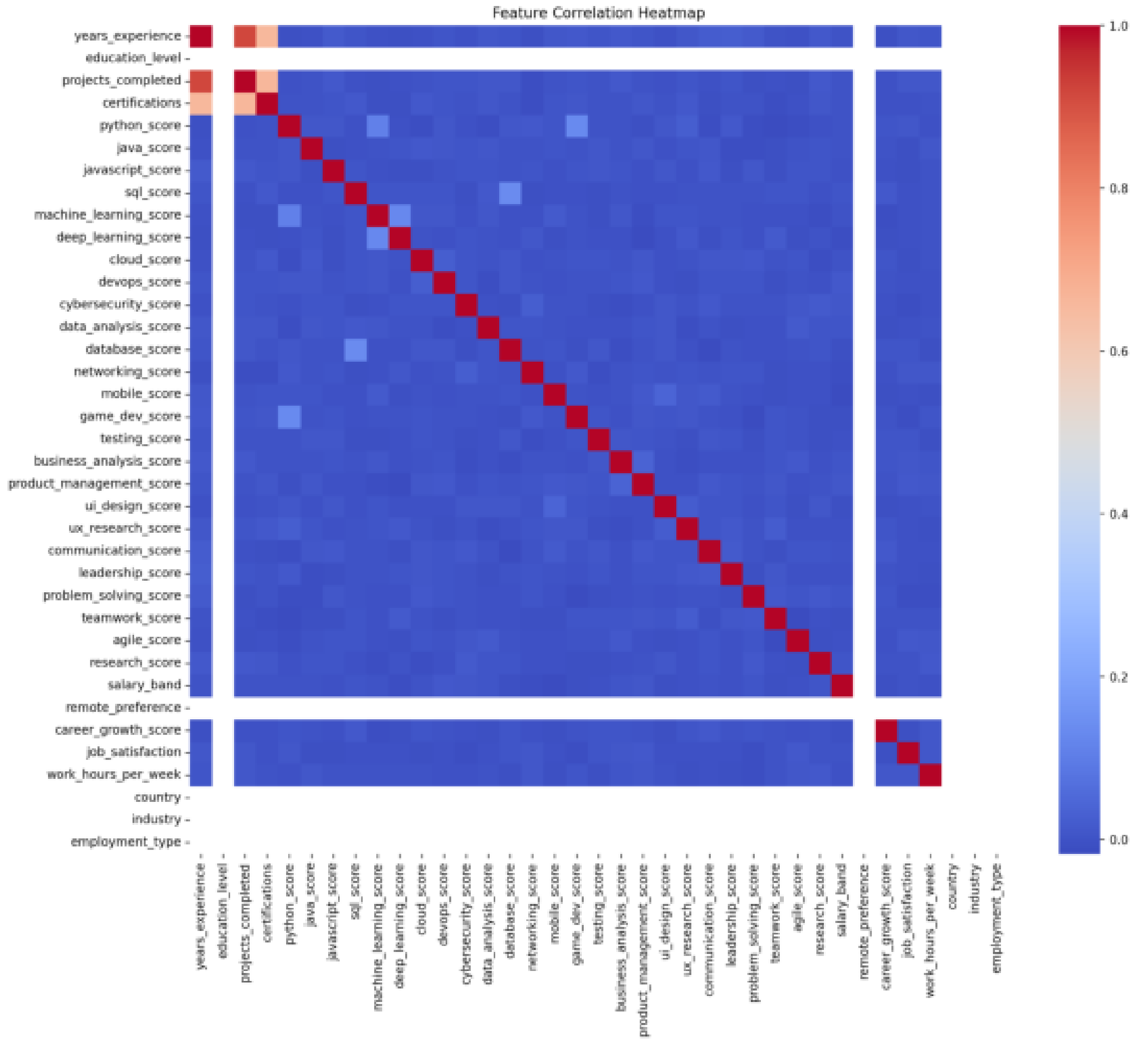
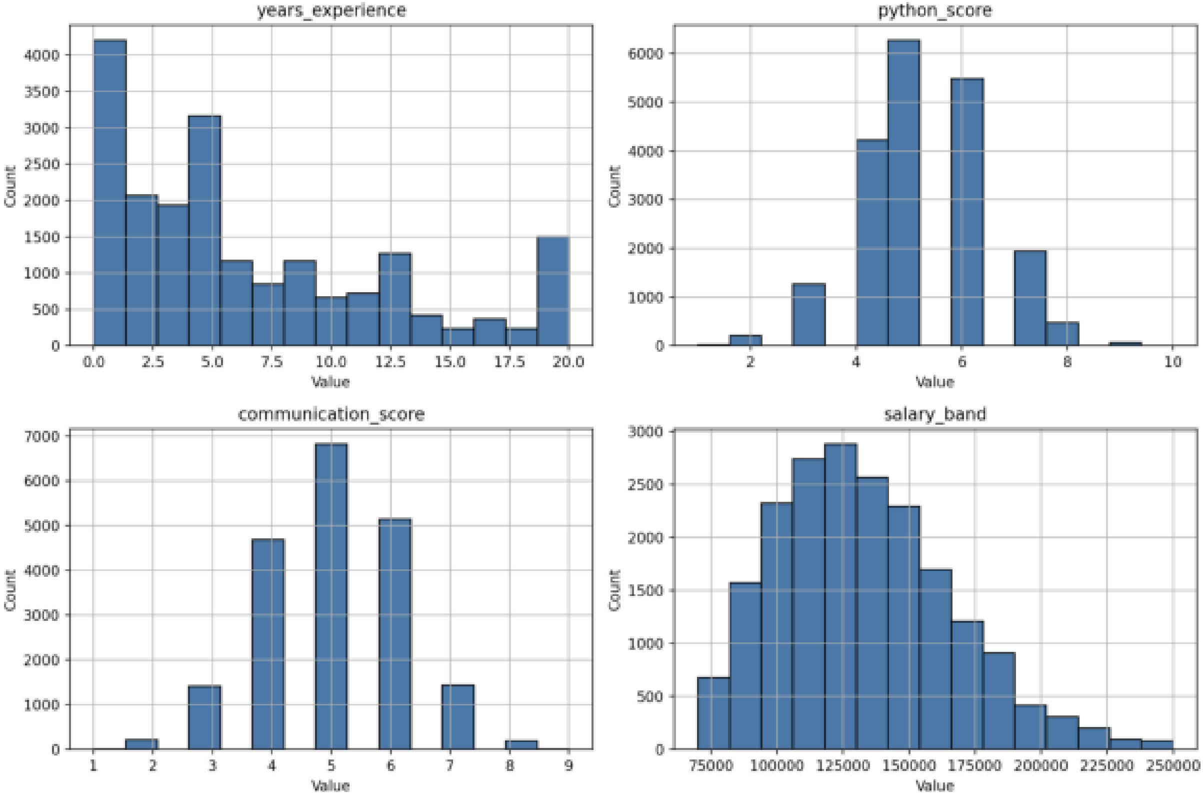
correlation_heatmap.png



distribution_plots.png